CE 3202 Concrete Mix Design (DoE method), Group G13–G16: filled values for pages 1–8

Given data: f_c = 35 MPa · slump 30–60 mm · SD = 3.1 MPa · 20 mm crushed coarse aggregate + uncrushed river sand · OPC · 45 MPa at w/c 0.5 · 35 % passing the 600 µm sieve · RD (SSD) = 2.65 · absorption: CA 0.43 %, FA 4.15 % · moisture: CA 0.39 %, FA 1.82 % · flexural loads P = 17.92, 18.89, 19.45 kN

Page 1: Selection of mix proportions

Target strength (F_t)	= F_c + 1.64 SD
SD	= 3.1 MPa
F_c	= 35 MPa
F_t	= 35 + 1.64 × 3.1 = 35 + 5.08
F_t	= **40.08 MPa ≈ 40.1 MPa**

From the graph of compressive strength vs free water/cement ratio (start at 45 MPa at w/c 0.5, follow the parallel curve down to 40.1 MPa):

- Water/cement ratio for 40.1 MPa = **0.55**

Free water content from the table of approximate free water contents needed for a **30–60 mm** slump (20 mm max. aggregate):

- Free water content for the uncrushed fine aggregate, W_f = **180 kg/m³**
- Free water content for the crushed coarse aggregate, W_c = **210 kg/m³**
- Free water content = ⅔ W_f + ⅓ W_c = ⅔(180) + ⅓(210) = 120 + 70 = **190 kg/m³**

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- Cement content = Free water content / water-cement ratio = 190 / 0.55 = **345.45 kg/m³ ≈ 345 kg/m³**
- Relative density of the aggregates (SSD) = **2.65**
- From the graph of free water content vs wet density of the concrete mix (190 kg/m³, RD 2.65): Wet density of concrete D = **2380 kg/m³**
- Total aggregate content = D − C − W
  - Cement content (C) = 345.45 kg/m³
  - Free water content (W) = 190 kg/m³
- Total aggregate content = 2380 − 345.45 − 190 = **1844.55 kg/m³**
- Fineness of the river sand, determined by the percentage passing the **600 µm** sieve, is found to be **35 %** of the total sand content.

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- Fineness fraction (proportion of fines, from the graph for 20 mm, slump 30–60 mm, w/c 0.55, 35 % passing) = **40 % (0.40)**
- Fine aggregate content = Total aggregate content × proportion of fines = 1844.55 × 0.40 = **737.82 kg/m³**
- Coarse aggregate content = Total aggregate content − Fine aggregate content = 1844.55 − 737.82 = **1106.73 kg/m³**

Table 2: Material requirement for 1 m³ (SSD basis)

Material	Amount (kg/m³)
Cement	345.45
Water	190.00
Coarse aggregate	1106.73
Fine aggregate	737.82

Page 4: Moisture adjustment (the aggregates are not in SSD condition)

Mass at natural moisture = SSD mass × (1 + MC) / (1 + absorption)

- Additional water from fine aggregates = Natural-state FA − FA content (SSD) = 737.82 × (1.0182 / 1.0415) − 737.82 = 721.31 − 737.82 = **−16.51 kg** (the sand is drier than SSD, so it absorbs 16.51 kg of water)
- Additional water from coarse aggregates = Natural-state CA − CA content (SSD) = 1106.73 × (1.0039 / 1.0043) − 1106.73 = 1106.29 − 1106.73 = **−0.44 kg** (the CA absorbs 0.44 kg of water)
- ∴ Water needed from outside = 190 + 16.51 + 0.44 = **206.95 kg/m³**
- Fine aggregate content (as weighed, natural moisture) = 737.82 × 1.0182 / 1.0415 = **721.31 kg/m³**
- Coarse aggregate content (as weighed) = **1106.29 kg/m³**

(Check: 345.45 + 206.95 + 721.31 + 1106.29 = 2380 kg/m³ ✓)

Page 5: Material requirements for 3 cubes, 3 prisms and the slump test

- Total volume of cubes (150 mm) = 3 × 0.15 × 0.15 × 0.15 = **0.010125 m³**
- Total volume of prisms (100 × 100 × 500 mm) = 3 × 0.1 × 0.1 × 0.5 = **0.015 m³**
- Volume of slump cone (h = 300, D = 200, d = 100 mm) = $\pi h/12\ (D^2 + Dd + d^2)$ = $\pi(0.3)/12 \times (0.04 + 0.02 + 0.01)$ = **0.0055 m³** (the same concrete is reused for the compaction factor test)
    - Total volume = 0.010125 + 0.015 + 0.0055 = **0.03062 m³**

- Volume of concrete mix with 15 % wastage = 0.03062 × 1.15 = **0.0352 m³ (≈ 0.035 m³)**

Table 03: Material requirements for 0.0352 m³ of concrete

Material	Amount (kg)
Water	206.95 × 0.0352 = **7.29**
Coarse aggregate	1106.29 × 0.0352 = **38.96**
Fine aggregate	721.31 × 0.0352 = **25.40**
Cement	345.45 × 0.0352 = **12.17**

Page 6: Compressive strength (150 mm cubes, A = 150 × 150 = 22 500 mm²)

Compressive strength = Failure load P / A

- Cube 1: compressive strength = P_1 (N) / 22 500 = ____ N/mm²
- Cube 2: compressive strength = P_2 (N) / 22 500 = ____ N/mm²
- Cube 3: compressive strength = P_3 (N) / 22 500 = ____ N/mm²

> Use your group's measured failure loads. (Example: 14 MPa corresponds to P = 14 × 22 500 = 315 kN.)

Page 7: Flexural strength (third-point loading, 100 × 100 mm section, span L = 300 mm)

- Average value of P = (17.92 + 18.89 + 19.45) / 3 = **18.75 kN**
- Value of P in newtons = 18.75 × 10^3 = **18 753 N**
- From the bending moment diagram: Maximum bending moment = PL / 6 = 18 753 × 300 / 6 = **937 667 N·mm (0.938 kN·m)**
- Second moment of area of the cross-section = bd^3 / 12 = 100 × 100^3 / 12 = **8.333 × 10^6 mm⁴**
- Flexural strength = (M / I) y = 937 667 × 50 / (8.333 × 10^6) = **5.63 N/mm²**

Page 8: Results

Slump	= ____ mm (measured)
Weight of the cylinder with partially compacted concrete	= ____ kg (measured)
Weight of the cylinder with fully compacted concrete	= ____ kg (measured)
Compaction factor	= partially compacted weight / fully compacted weight = ____
Compressive strength	= ____ MPa (average of the 3 cubes, ≈ 14 MPa)
Flexural strength	= **5.63 N/mm²**

Assumptions to check against your lecture notes: the values read from the DoE graphs (w/c 0.55, D = 2380 kg/m³, fines 40 %); 150 mm cubes; 100×100×500 mm prisms on a 300 mm span. These sizes give the ≈0.035 m³ batch volume stated in the lab sheet.